

Detailed Feature Engineering Planning Report

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Introduction

Feature engineering is one of the most important stages in data science because raw healthcare data often lacks directly predictive variables. The objective of this planning report is to identify meaningful derived features for patient recovery prediction and diet recommendation.

BMI Feature

BMI will be calculated using patient weight and height. BMI helps identify obesity risk, nutritional imbalance, and overall health status. This feature is expected to improve diet recommendation quality.

Recovery Score

A recovery score will be engineered using pain levels, sleep quality, energy levels, medication adherence, and symptom severity. This score can be used as a target variable for recovery prediction models.

Compliance Score

Compliance metrics will measure how consistently patients follow medication schedules, meal plans, hydration goals, and exercise recommendations.

Risk Classification Features

Risk categories will be derived using blood pressure, blood sugar levels, heart rate, and clinical symptoms to classify patients into low, medium, or high-risk groups.

Activity-Based Features

Daily steps, exercise type, and energy levels will be combined to create activity intensity indicators useful for wellness analysis.

Temporal Features

Tracking dates will be transformed into day-based progression metrics to study recovery improvement trends over time.

Machine Learning Importance

The engineered features will improve pattern detection, enhance predictive accuracy, reduce noise, and support healthcare recommendation systems.

Planned ML Applications

Engineered features will support regression, classification, recovery prediction, patient risk analysis, and personalized diet recommendation systems.